EDI – SMARTCELL

Activity 2- : Understand the imaging conditions

1. Table: - Malaria (Infected)

- **ID – IMG_001**

Parameter	Test 1	Test 2	Final
Magnification	100× (provided setting)	400× (provided setting)	Actual magnification unverified
Illumination	Low (test condition)	Medium (test condition)	Appears moderately illuminated
Focus	Poor (test condition)	Good (test condition)	Slightly soft; fine details unclear
Resolution	720p (test condition)	1080p (test condition)	Original resolution unverified
Camera	USB microscope + laptop camera	USB microscope (test condition)	USB microscope reported;
Distance from camera to eyepiece	Not provided	Not provided	Unknown
Background	Light pink, with black surrounding areas	—	Uneven, with dark borders
Staining	Pink coloration visible	—	Stain type unknown
Image blur	Mild softness	—	Fine details are not sharply resolved
Shadows	Dark areas around the edges	—	Present around the displayed image
Uneven lighting	Slight variation in brightness	—	Appears mildly uneven
Slide type	Not provided	—	Unknown
Image quality	Moderate	—	Fair; limited fine detail
Remarks	Broad, pale pink oval region	—	Labeled “infected” in the dataset; biological status cannot be independently confirmed

2. Table: - Malaria (Parasitized)

- **ID – IMG_002**

Parameter	Test 1	Test 2	Final
Magnification	100× (provided setting)	400× (provided setting)	Actual magnification unverified
Illumination	Low (test condition)	Medium (test condition)	Appears moderately illuminated
Focus	Poor (test condition)	Good (test condition)	Soft focus
Resolution	720p (test condition)	1080p (test condition)	Original resolution unverified
Camera	USB microscope + laptop)	USB microscope (test condition)	USB microscope detected.
Distance from camera to eyepiece	Not provided	Not provided	Unknown
Background	Pink with black surrounding borders	—	Uneven
Staining	Pink and darker purple areas	—	Stain type unknown
Image blur	Visible softness	—	Fine structures are difficult to distinguish
Shadows	Dark edge regions	—	Present
Uneven lighting	Brightness varies across the image	—	Appears uneven
Slide type	Not provided	—	Unknown
Image quality	Fair to poor	—	Limited detail due to blur
Remarks	Elongated, darker purple structure visible within the pink region	—	Labeled “parasitized” in the dataset; the structure cannot be confirmed as a parasite from this photograph alone

3. Table: - Malaria (Uninfected)

- **ID – IMG_003**

Parameter	Test 1	Test 2	Final
Magnification	100× (provided setting)	400× (provided setting)	Actual magnification unverified
Illumination	Low (test condition)	Medium (test condition)	Appears moderately illuminated
Focus	Poor (test condition)	Good (test condition)	Soft focus
Resolution	720p (test condition)	1080p (test condition)	Original resolution unverified
Camera	USB microscope + laptop	USB microscope (test condition)	USB microscope detected
Distance from camera to eyepiece	Not provided	Not provided	Unknown
Background	Pale grey-pink with black surrounding borders	—	Uneven
Staining	Pale pink-purple coloration	—	Stain type unknown
Image blur	Visible softness	—	Fine details are not clearly resolved
Shadows	Dark regions around the image boundary	—	Present
Uneven lighting	Mild brightness variation	—	Appears mildly uneven
Slide type	Not provided	—	Unknown
Image quality	Fair	—	Broad outline visible, but fine details are limited
Remarks	Large, pale oval region with indistinct internal detail	—	Labeled “infected” in your dataset; infection cannot be independently confirmed

4. Table: - ALL (Infected-cell region marked in blue)

- **ID – IMG_004**

Parameter	Test 1	Test 2	Final
Magnification	100× (provided setting)	400× (provided setting)	Actual magnification unverified
Illumination	Low (test condition)	Medium (test condition)	Appears bright, with some pale areas
Focus	Poor (test condition)	Good (test condition)	Soft focus
Resolution	720p (test condition)	1080p (test condition)	Original resolution unverified
Camera	USB microscope + laptop	USB microscope (test condition)	USB microscope deteced
Distance from camera to eyepiece	Not provided	Not provided	Unknown
Background	Light pink-grey	—	Textured, with pale regions
Staining	Blue-purple and pink coloration	—	Stain type unknown
Image blur	Visible softness	—	Boundaries and internal features are indistinct
Shadows	Mild dark variation near some boundaries	—	Some tonal variation
Uneven lighting	Brightness varies slightly	—	Appears mildly uneven
Slide type	Not provided	—	Unknown
Image quality	Fair	—	Multiple blue-purple regions visible, but details are limited
Remarks	Several overlapping or closely positioned blue-purple structures	—	Blue regions designated as infected in your dataset; ALL status is not independently verifiable from the image

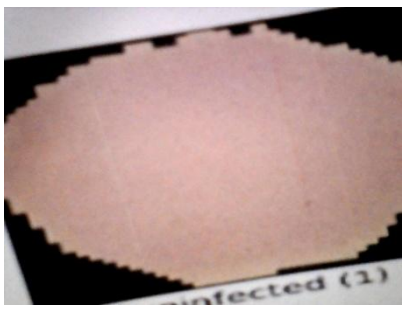
5. Table: - ALL (Infected-cell region marked in blue)

- **ID – IMG_005**

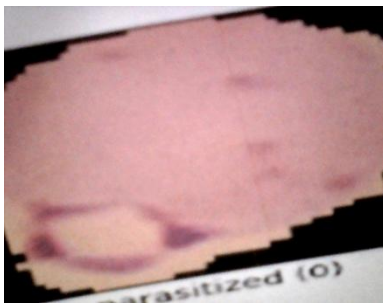
Parameter	Test 1	Test 2	Final
Magnification	100× (provided setting)	400× (provided setting)	Actual magnification unverified
Illumination	Low (test condition)	Medium (test condition)	Appears moderately bright
Focus	Poor (test condition)	Good (test condition)	Soft focus
Resolution	720p (test condition)	1080p (test condition)	Original resolution unverified
Camera	USB microscope + laptop	USB microscope (test condition)	USB microscope detected
Distance from camera to eyepiece	Not provided	Not provided	Unknown
Background	Pale grey-pink	—	Textured
Staining	Blue-purple coloration	—	Stain type unknown
Image blur	Visible softness	—	Edges are not sharply defined
Shadows	Minor tonal variation	—	Some darker areas within the blue regions
Uneven lighting	Mild brightness variation	—	Appears mildly uneven
Slide type	Not provided	—	Unknown
Image quality	Fair	—	Two prominent blue-purple regions visible; internal details are limited
Remarks	Two large blue-purple structures against a pale background	—	Blue regions designated as infected in your dataset; ALL status is not independently verifiable from the image

Activity 3- : Image acquisition records

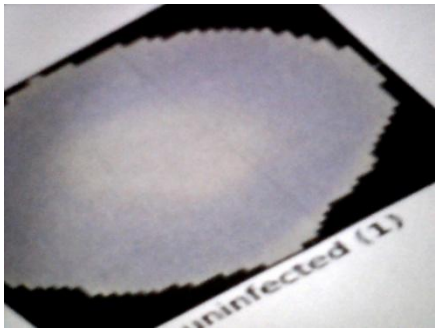
Image ID	Date	Magnification	Camera	Lighting	Slide type	Image quality	Remarks
IMG_001	11/10/26	Unknown	USB microscope reported; + Laptop camera	Moderate, slightly uneven illumination	Printed image	Fair	Malaria - labeled infected
IMG_002	11/10/26	Unknown	USB microscope reported; + Laptop camera	Moderate illumination, slightly uneven	Printed image	Fair to poor	Malaria - labeled parasitized
IMG_003	11/10/26	Unknown	USB microscope reported; + Laptop camera	Moderate, slightly uneven illumination	Printed image	Fair	Malaria - labeled infected
IMG_004	11/10/26	Unknown	USB microscope reported; + Laptop camera	Adequate illumination with visible background texture	Printed image	Fair	ALL - blue regions labeled infected
IMG_005	11/10/26	Unknown	USB microscope reported; + Laptop camera	Moderate illumination with slight brightness variation	Printed image	Fair	ALL - blue regions labeled infected



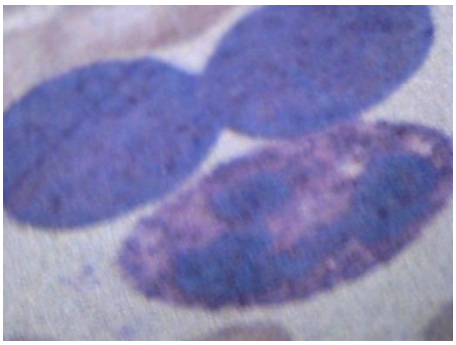
IMG_001



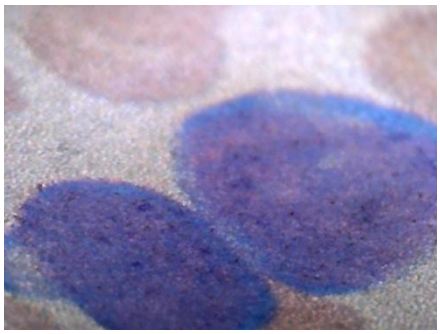
IMG_002



IMG_003



IMG_004



IMG_005

Activity 5: Image Quality Analysis

- **Summary of image quality**

Image ID	Category	Main issue	Overall quality
IMG_001	Malaria — infected	Mild blur	Fair
IMG_002	Malaria — parasitized	Blur and indistinct details	Fair to poor
IMG_003	Malaria — infected	Mild blur	Fair
IMG_004	ALL	Soft boundaries and visible texture	Fair
IMG_005	ALL	Soft boundaries and visible texture	Fair

❖ **IMG_001 - Malaria**

- ✓ Illumination acceptable
- ✗ Slightly blurry
- ✓ Main region visible
- ✓ No major reflection
- ✗ Fine details unclear

❖ **IMG_002 - Malaria**

- ✓ Illumination acceptable
- ✗ Blurry; focus needs improvement
- ✓ Main structures visible
- ✓ No major reflection
- ✗ Fine details unclear

❖ **IMG_003 - Malaria**

- ✓ Illumination acceptable
- ✗ Slightly blurry
- ✓ Main region visible
- ✓ No major reflection
- ✗ Fine details unclear

❖ **IMG_004 - ALL**

- ✓ Coloured regions visible
- ✗ Boundaries slightly blurry
- ✓ Illumination acceptable
- ✗ Some image texture/noise
- ✓ No major reflection

❖ **IMG_005 - ALL**

- ✓ Coloured regions visible
- ✗ Boundaries slightly blurry
- ✓ Illumination acceptable
- ✗ Some image texture/noise
- ✓ No major reflection